

Prevalence of Depression and Anxiety in College Students

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Abstract

Context: Depression and anxiety are the most common mental disorders existing today among the youngsters owing to academic excellence pressure and societal living expectations. They are overlooked most of the times and can have a grave impact on a student's personal, academic, and social life. **Aim:** The aim of the study was to find the prevalence of depression and anxiety in college students. **Methods:** A prospective study was done in 400 college students in the age group of 18–25 years. A self-made questionnaire, along with the Beck's Depression and Anxiety Inventory, was administered. **Results:** We found that depression was prevalent in about 48.30% of the students, whereas anxiety was prevalent in 50% of these students. **Conclusion:** Hence, moderate-to-severe levels of depression and anxiety exist in college students, which need to be confirmed and diagnosed clinically and treated at the earliest.

Keywords: Anxiety, college students, depression

INTRODUCTION

The World Health Organization (WHO) defines health as “A state of complete physical, mental and social wellbeing and not merely the absence of disease or infirmity.”^[1]

Mental health is still considered a stigma, and millions of them are silently suffering from it without any external help or support.^[2]

Adolescents, especially college-going students, suffer from both depression and anxiety at higher rates as the stressors/triggers are present in abundance. Emotional, behavioral, sexual, economic, academic, and social changes and efforts of discovering one's identity with psychosocial and maturation also occur.^[3] They go through critical transitory period in their life in which they are going from adolescence to adulthood making major life decisions.^[4] During this period, the mental health of university youth constitutes one of the important components of social health.^[5] It is known to affect an individual's academic performance greatly. According to the WHO report, practically, all psychiatrists who have had extensive experience in working with college students agree that about 10% of the members of any institution of higher learning are likely to have emotional problems at some time or other during each year which interferes seriously with their work.^[1] Depression is underrecognized among adolescents because depressive symptoms are considered a familiar part of

adolescent experience.^[6] These stressors often exert a negative effect with catastrophic consequences on students' academic performance, physical health, and psychological well-being.^[7]

This study aims to evaluate the prevalence of depression and anxiety in college students.

METHODS

Ethical clearance – The study was approved and ethical clearance was given by the Ethical Committee of School of Physiotherapy at D. Y. Patil University, Nerul, Navi Mumbai. Furthermore, informed consent was obtained before subjecting the participant in the study.

Study design – It was a prospective-based study. A sample of 400 students from various colleges of D. Y. Patil Deemed to be University, Nerul, Navi Mumbai, Maharashtra, were selected.

The students belonged to various schools of the same university such as D. Y. Patil School of Medicine (medicine

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students), D. Y. Patil School of Physiotherapy (physiotherapy students), and D. Y. Patil School of Dentistry (dentistry students) and RAIT Institute (engineering students). Out of 400 students, 200 of them were from the medical colleges, whereas the rest of them from the engineering institute. Data were collected with the help of a self-made questionnaire consisting of sociodemographic data and the Beck's Depression Inventory (BDI) and Beck's Anxiety Inventory (BAI).^[8] The forms were majorly distributed in the form of hard copies, whereas some were forwarded digitally through Google Forms. For reliability and anonymity, confidentiality was maintained. The following questionnaires were used in the study to determine the prevalence of depression and anxiety:

1. BDI (Beck *et al.*, 1961) – It is a 21-item self-reporting scale which examines characteristics, attitudes, and symptoms of depression. Internal consistency for the BDI ranges from 0.73 to 0.92, with a mean of 0.86.^[6] The BDI demonstrates high internal consistency, with alpha coefficients of 0.86 and 0.81 for psychiatric and nonpsychiatric populations, respectively.^[9] It evaluates key symptoms of depression including mood, pessimism, sense of failure, self-dissatisfaction, guilt, punishment, self-dislike, self-accusation, suicidal ideas, crying, irritability, social withdrawal, indecisiveness, body image change, work difficulty, insomnia, fatigability, loss of appetite, weight loss, somatic preoccupation, and loss of libido
2. BAI (Beck *et al.*, 1993) – It is a 21-item scale that showed high internal consistency ($\alpha = 0.92$) and test-retest reliability over 1 week, $r(81) = 0.75$. The BAI was moderately correlated with the revised Hamilton Anxiety Rating Scale (0.51) and mildly correlated with the Hamilton Depression Rating Scale (0.25) (Beck *et al.*, 1988 and 1993). It is a four-point rating scale which consists of 21 adjectives related to different domains of anxiety symptoms such as cognitive, physiological, emotional, and behavioral.^[10]

Both the inventories are not diagnostic in nature; rather, they help in knowing the severity of depression and anxiety in the general population. However, some studies have proved that they can be used as a diagnostic tool in community setup.^[11]

Inclusion criteria

Undergraduate, graduate, and postgraduate students who were in the adolescent age group of 18–25 years and who had been exposed to the curriculum for at least 1 year were included in the study.

Exclusion criteria

Students who were unwilling to participate and who submitted incomplete data and those who did not come under the given inclusion criteria were excluded from the study.

Microsoft Excel was used for calculating the findings and results.

RESULTS

About 400 college students, belonging to the age group of 18–25 years, participated in our study. Out of them, 63.5% were female, whereas 36.5% were male. Figures 1 and 2 depict the levels of depression and anxiety, respectively. Our results show that about 26.5% of the college students were suffering from depression at clinically significant levels and about 6.14% of them had potentially concerning level of anxiety. According to the grading of BDI,^[12] 9% of the students had borderline clinical depression which requires clinical attention.

DISCUSSION

In general, depression was found to be prevalent in about 48.30% of the students, whereas anxiety was prevalent in 50% of these students. Out of these, 26.75% of them were between borderline and extreme stage of depression which requires clinical attention. These findings were consistent with other similar studies. A study was conducted among the students of the Punjab University, Chandigarh in 2014, which showed the overall prevalence of depression, anxiety, and stress to be 59.3%, 86.5%, and 52.8%, respectively, higher than that of ours,^[2] whereas a study carried out in Karnataka in 2012 depicted an overall prevalence of depression to be 71.25%. Among those, a majority (80%) had mild and moderate degree of depression. The prevalence of severe and profound depression was 7.5% and 6.7%, respectively.^[13] Furthermore, a study conducted in Jhansi, Uttar Pradesh, in the year 2016 revealed that depression was 57% prevalent, whereas anxiety was 71%. Among those with depression, a majority (73%) had mild and moderate degree of depression. Only 3% of the respondents had extremely severe depression, whereas 67% of the anxious respondents had mild and moderate degree and 29% had severe degree of anxiety.^[14] A study carried out in a Turkish university in 2008 revealed that depression, anxiety, and stress levels of moderate severity or above were found in 27.1%, 47.1%, and 27% of their respondents, respectively,^[15] whereas a study carried out in 2006 in a Pakistani university showed a very high prevalence (70%) of depression and anxiety in the students.^[16]

Studies have reported wide variations in the proportion of students identified as depressed, from relatively low rates around 10% to high rates of between 40% and 84%.^[17]

The percentages, though less, are of grave importance in terms of its effect on the students' quality of life and its future consequences. This shows that there is an urgent and extreme requirement of identifying and treating the depression as earliest as possible.

College students have to undergo a lot of changes in their personal, professional, and extracurricular life. This necessitates a lot of stressful situations that they need to face as a part of their growing process.

In India, there are various important factors associated with depression in adolescents including gender, low level of

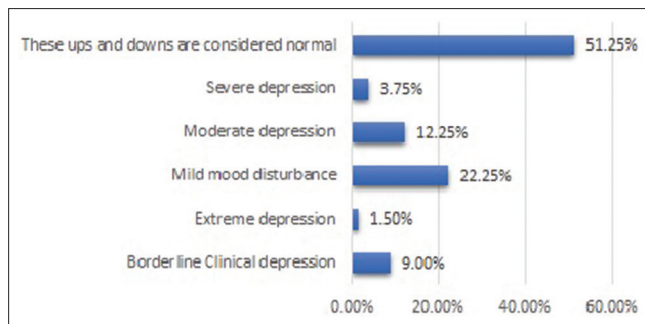


Figure 1: Levels of depression in college students according to the grading of Beck's Depression Inventory. It shows that out of 400 students, 22.25% of them have mild mood disturbances, 9% of them have borderline clinical depression, 12.25% of them have moderate depression, whereas 3.75% and 1.5% of students have severe and extreme levels of depression, respectively

parental warmth, loneliness, criticism escalating adolescent–parent conflict; socioeconomic status. In addition to this, parental expectations, perceived rejection by peers, parents, and teachers, academic excellence pressure, fear of failure, lack of supportive environment in the university, favoritism, relationships, unmatched deadlines predict increase in depressive symptoms.^[18] Depression and severe suicidal ideation are also linked to being bullied or to acting as bullies. Lifestyle is another important issue, as factors indicative of the adoption of nontraditional lifestyle are associated with an increase in the prevalence of depression.^[19]

All these factors in some or the other way contribute in adding to the stress faced by these students. There are some students that require help in facing these problems but are not diagnosed until it becomes extremely severe. This can many a times have a potentially dangerous outcome.^[19]

The high rates of depression, anxiety, and stress among university students have major implications, not only with psychological morbidity that will have adverse effects on students' health, development, educational attainment, and quality of life but also the deteriorating influence on their own families, institutions, and even on other people's lives. Hence, it is necessary to identify and take appropriate actions as early as possible.^[15]

The ways in which students choose to cope with the stressors of their training will eventually act as the blueprints for how they will deal with future professional and personal stresses.^[7] Academic factors are found to be the most important stressors. Hence, there is a pressing need for definite measures to decrease substantially the burden of stress on the students.

Depression, anxiety, stress, and substance abuse are mere symptoms of the hidden, unresolved, and ignored emotional issues which cumulatively grow inside insidiously.^[2] Universities are well positioned to promote mental health among young people because they encompass several important aspects of students' lives: academics, health services, residences, social networks, and extracurricular activities.^[19] Very few institutions

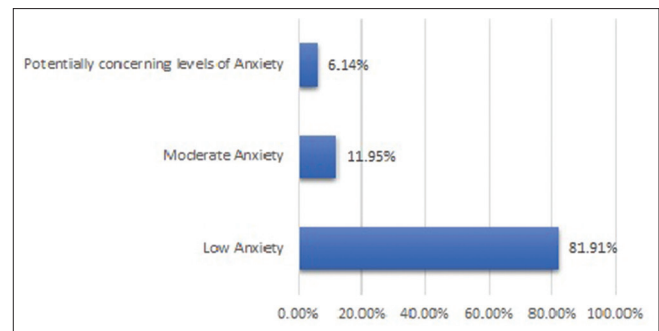


Figure 2: Levels of anxiety in college students according to the grading of Beck's Anxiety Inventory. It denotes that out of 400 college students, 81.91% have low levels of anxiety, 11.95% of them have moderate levels of anxiety, whereas 6.14% of them have potentially concerning levels of anxiety

have sufficiently well-developed psychiatric services to deal with that number of students.

CONCLUSION

Hence, this study helps in determining the need for early scanning with simple questionnaires to identify the severity of depression and anxiety prevalent in college students. Universities are well positioned to promote mental health among young people because they encompass several important aspects of student's lives : academics, health services, residencies, social networks, and extracurricular activities.^[19]

Thus they need to equip themselves in order to take the necessary measures for the prevention or treatment of depression and anxiety in their students.

Limitations

Our study had a limited small sample size. The study was based on a single university, and other specialties of college students other than the ones mentioned were not included. As mentioned earlier, the scales were distributed and the information collected was subjective to the individuals. No formal interviews were conducted. Furthermore, both BDI and BAI are self-reported measures; therefore, they cannot be completely relied on. The symptoms shown by the participant may differ from the ones they marked upon. The big disadvantage is the ability for nonmotivated patients to manipulate the process and state what they feel will lead to them receiving a diagnosis of depression and having some type of personal gain.^[20] Therefore, a proper psychiatric evaluation is necessary of these kinds of individuals, which is not carried out in this study.

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We assure that the manuscript has been read and approved by all the authors, that the requirements for authorship have been met, and that each author believes that the manuscript represents honest work.

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Conflicts of interest

There are no conflicts of interest.

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